Keywords: test-time adaptation • robot locomotion • reinforcement learning • distribution shift

TEMPO: Efficient Test-Time Adaptation For Robot Policies

Critic Entropy Regularization and Score-Based Memory Enable On-Device Policy Adaptation—Up to 80% Performance Recovery on Physical Quadrupeds Under Load

By Motasem Alfarra, Pietro Mazzaglia, Markus Peschl, Daniel Dijkman, Christos Louizos

arXiv:2610.20318 • NeurIPS 2026

Reinforcement-learned locomotion policies trained in simulation degrade in deployment when conditions shift—payload changes the center of mass, terrain modifies effective friction, sensors drift with temperature. TEMPO addresses this with a surprisingly effective combination: an entropy regularizer on the critic that prevents value collapse under distribution shift, paired with a score-based memory buffer that upweights hard observations during on-device adaptation. Across three quadruped platforms (Go1, Go2, G1) and four shift types in MuJoCo Playground, TEMPO yields up to 80% improvement in the single-device setting; on a physical Unitree Go2 carrying an added payload, TEMPO recovers pre-shift performance within minutes.

AT A GLANCE

Problem: Sim-to-real locomotion policies degrade when deployment conditions (payload, terrain, sensors) differ from training—and no simulator is available at test time.

Why it matters: Real-world deployment requires policies that adapt autonomously; simulator-based fine-tuning is impractical on hardware.

Method: Critic entropy regularizer prevents value collapse; score-based memory oversamples observations the policy finds hard.

Result: Up to 80% performance improvement in simulation; physical Unitree Go2 recovers from payload shift within 10 minutes.

Evidence: Go1, Go2, G1 in MuJoCo Playground under four distribution shifts; physical Unitree Go2 experiment.

THE BIG PICTURE

Reinforcement learning in simulation remains the dominant approach for robot locomotion, but the sim-to-real gap means deployed policies face conditions their training never covered. Once deployed, the robot's own experience is the only data available for further improvement.

Test-time adaptation (TTA) of neural networks has been studied for supervised classification, but RL policies present a harder challenge: no ground-truth labels, expensive reward evaluation, critic divergence under distribution shift, and compute constrained to a single embedded device.

WHY EXISTING APPROACHES FALL SHORT

Standard PPO fine-tuning at test time: Without a simulator, PPO must use on-device rollouts directly. Distribution shift causes the critic to diverge, destabilising policy updates.

Domain randomisation: Broad randomisation reduces peak performance and still fails for shifts outside the randomised range.

Meta-learning (MAML, PEARL): Requires a distribution over adaptation tasks at training time—unavailable when the real-world distribution is unknown—and inner-loop gradient computation expensive on embedded hardware.

CORE MECHANISM

1. Critic Entropy Regularizer. The critic loss is augmented with an entropy penalty: L_V = L_TD + beta * H(V_phi(s)), where H(V_phi(s)) is the entropy of the value distribution across a replay batch. This prevents the critic from collapsing to a near-constant value function under distribution shift.

2. Score-Based Memory. Each observation s in the replay buffer is assigned a score proportional to its TD error: score(s) proportional to |r + gamma*V(s') - V(s)|. High-scoring observations (where the critic is most wrong) are sampled more frequently, biasing updates toward the hardest parts of the new distribution.

TECHNICAL FORMULATION

Full TEMPO objective at each adaptation step:

L_TEMPO = L_PPO + lambda_V [L_TD + beta * H(V_phi)]

Score-based sampling probability for transition i:

p_i = score(s_i)^alpha / sum_j score(s_j)^alpha, alpha = 0.6

(Priority exponent follows the PER convention.)

Entropy regularizer guarantee: H(V_phi(s)) >= H_min > 0 for all t, ensuring policy gradient variance stays non-zero throughout adaptation.

LEARNING OR INFERENCE PROCEDURE

Offline training (simulation): Standard PPO with domain randomisation. No modifications to the training procedure.

Online adaptation (deployment): At each step: (1) execute policy, collect transitions; (2) update transition scores from critic TD errors; (3) sample batch from priority queue; (4) apply one PPO + entropy-regularised critic update. Total compute per step equivalent to one standard PPO update.

WHAT THE GUARANTEE SAYS

The entropy regularizer prevents critic collapse: H(V_phi(s)) >= H_min > 0 for all t, ensuring policy gradient variance stays non-zero. This provides a necessary condition for adaptation progress; convergence rate depends on the magnitude of distribution shift and data collection speed.

EXPERIMENTAL FINDINGS

Go2 (MuJoCo Playground, payload shift): TEMPO reaches 80% of pre-shift performance within 5 minutes; standard PPO fine-tuning diverges.

Go1, G1 (four shift types): Consistent gains; average +42% over non-adapted baseline.

Physical Unitree Go2: TEMPO with 2 kg added payload recovers stable locomotion within 10 minutes; non-adapted policy fails to sustain a trot gait.

Baseline comparison: Outperforms domain randomisation, meta-RL (MAML), and test-time normalisation across all conditions.

ABLATIONS AND INTERPRETATION

Entropy regularizer only: Without score-based memory, adaptation is slower (-25% performance at 5 min); both components are needed.

Score-based memory only: Without entropy regularization, the critic still diverges in 3 of 4 shift scenarios.

Priority exponent alpha: Performance robust to alpha in [0.4, 0.7]; alpha < 0.4 approaches uniform sampling and loses most gains.

Batch size: Adaptation improves up to batch size 64 and plateaus.

REFERENCE

Motasem Alfarra, Pietro Mazzaglia, Markus Peschl, Daniel Dijkman, Christos Louizos. **Efficient Test-Time Adaptation For Robot Policies.** arXiv:2610.20318. NeurIPS 2026. https://arxiv.org/abs/2610.20318